

Anshu Arora

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Education

- Stanford University**, MS in Electrical Engineering, specialization in AI and Optimization Sep 2026 – Ongoing
- **Key courses:** Robot Perception, Robot Autonomy-I, Decision Making Under Uncertainty, Markov Decision Processes
- IIT Bombay**, B. Tech in Electrical Engineering Oct 2022 – May 2026
- GPA: **9.43/10.0**, with **Honors** in Electrical Engineering and a **Minor** in Data Science and Artificial Intelligence
 - **Key courses:** Distributed Optimization and Learning, Advanced Machine Learning, Mathematics of Deep Learning, Online Learning, Wavelets, Control Systems, Probability and Random Processes, Data Structures and Algorithms

Publications

- **Anshu Arora**, Satish Mulleti, "A Lowrate Variable-Bias Integrate-and-Fire Time Encoding Machine", published IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), April 2025
- **Co-author**, "Technical Design Report of Matsya 6D, Autonomous Underwater Vehicle", Naval Engineers Journal, Vol. 136, pp. 161–175, September 2024
- K Yashaswini*, **Anshu Arora***, Satish Mulleti, "A Non-Uniform Quantization Framework for Time-Encoding Machines", under review for IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026
- **Anshu Arora***, K Yashaswini*, Satish Mulleti, "Linear-Bias Time Encoding for Low-Rate Quantized Representation of Bandlimited Signals", currently under review for IEEE Signal Processing Letters **equal contribution*

Scholastic Achievements

- Selected as a recipient of **National University of Singapore Fellowship** for demonstrating research excellence 2025
- Received the **Undergraduate Research Award** for work in signal processing and machine learning 2025
- **Ranked 233** (top 0.02%) in the **JEE Advanced** and **917** (top 0.09%) in the **JEE Mains** (1M+ candidates) 2022
- Awarded the **KVPY Fellowship** conducted by the Department of Science and Technology, **Govt. of India** 2022

Research Projects

- Clustering in Federated Reinforcement Learning** Mar 2025 – May 2026
Guides: Prof. Avishek Ghosh and Prof. Pranay Sharma, Computer Science and Engineering, IIT Bombay
- Conducting an in-depth literature review on **federated reinforcement learning** with **TD(0)** and **linear function approximation**, focusing on clustering approaches for **policy evaluation** across distributed agents
 - Designed a novel algorithm and rigorously derived **convergence bounds** using a **one-shot clustering approach** followed by collaboration of agents in a cluster through aggregation of local TD(0) update directions
 - Analyzed the trade-off between misclassifications of agents and **linear speed-up** in convergence due to collaboration
- Robustness in Federated Learning** Sep 2025 – Dec 2025
Guide: Prof. Pranay Sharma, Center for Machine Intelligence and Data Science, IIT Bombay
- Investigating the existing research on Byzantine agents and robust aggregation methods in multi-agent systems
 - Modeling the notion of **stealth in adversarial attacks**, as a multi-objective optimization problem
 - Evaluating the attack success rate using the **MGDA algorithm**, to converge to **Pareto stationary** points
- Consensus on Lie Groups with Unreliable Networks** Jun 2025 – Dec 2026
Guide: Prof. Ravi Banavar, Systems and Control Engineering, IIT Bombay
- Examining consensus dynamics on **Lie Groups** under lossy connectivity in **multi-agent information networks**
 - Analyzing the relation between the **Vicsek model** and **first-order kinematic model** of consensus, employing a **switching function analysis** and local strong convexity assumptions on Lie groups with a **bi-invariant metric**
 - Characterizing the **domain of initial conditions** that permit consensus despite communication disruptions
- Time Encoding Machines (TEMs)** May 2024 – Nov 2025
Guide: Prof. Satish Mulleti, Electrical Engineering, IIT Bombay
- Designed a time-varying bias to closely track the input signal, leading to a significant **reduction in oversampling**
 - Derived **mathematical transformations** between the probability density function of the amplitude and time encodings, to design an efficient **non-uniform quantizer**, significantly reducing the information throughput
 - **Bypassed the Bernstein inequality** in the operator-based reconstruction mechanism of TEMs, to remove the

bottleneck of sampling above the Nyquist rate, without compromising on the perfect reconstruction guarantee

Key Projects

Classification using Spiking Neural Networks (SNNs)

Oct 2025 – Dec 2025

- Used SNNs with **spike time dependent plasticity** as the learning mechanism to perform unsupervised **MSNIST dataset classification**, based on the Leaky Integrate-and-Fire neuron model architecture
- Developed a supervised MNIST classifier using **snnTorch** with **backpropagation through time training**

Inference in Machine Learning

Jan 2025 – May 2025

- Evaluated **message passing algorithms** for computing marginals and MAP assignments on graphical models
- Implemented **denoising diffusion probabilistic models** for unconditional and conditional generation, analyzing the effects of diffusion steps, noise schedules, and classifier-free guidance on sample quality
- Deployed **MCMC sampling algorithms** for energy-based models and implemented **Gaussian Process regression**

EMG Controlled Prosthetic Arm

Jan 2025 – May 2025

- Placed **top 3** among **40+ teams** for developing a myoelectric system to classify **3 motions** from EMG signals
- Applied signal processing for **feature extraction** and implemented an **SVM-based model** on Neuropype software

Optimization of Traffic Light Timings

Nov 2024 – Dec 2024

- Built **linear programming models** on a time-space network with **Pyomo** to minimize the commuter wait times
- Analysed **reinforcement learning algorithms** to optimize timings under stochastic intersection conditions

Wavelets in Magnetic Resonance Imaging

Jul 2024 – Nov 2024

- Designed a **CNN-based deep learning architecture** using **curvelets** as features for tri-segmentation of the brain
- Incorporated **high-resolution time-frequency features** into the model, achieving an **IoU score of 0.71**

Positions of Responsibility

Mentorship Program Head, Electrical Dept. – Student Mentorship Program

March 2025 – Mar 2026

- One of 3 departmental student representatives leading **1K+ students**, entrusted with advancing academic excellence
- Leading a **3-tier mentorship network** of 5 subgroups and **50+ mentors** to support **200+ undergraduates**

Captain, Women's Basketball Team – IIT Bombay Sports

Aug 2024 – Aug 2025

- Led the team to its **first ever Gold Medal** at the 57th Inter-IIT tournament, from **20+** participating IITs
- Coordinating fitness initiatives, structured team practice schedules, fostering **strong team cohesion** and bonding

Chief Business Executive – Autonomous Underwater Vehicle Team

Aug 2025 – May 2026

- Part of a **50+ member student robotics team** developing **Matsya**, an autonomous underwater vehicle integrating visual, inertial, acoustic, and depth sensing with thruster and servo actuation for realistic naval missions
- Leading the business division, responsible for procuring **sponsorships** and performing outreach activities
- Achieved an overall **4th rank** in the international competition **RoboSub '24** at California among **40+ teams**

Experiences

Volunteering Activities

- Volunteered with **Ummeed NGO**, teaching mathematics to disadvantaged students over the summers 2024
- Worked with young girls to take up STEM as a part of **Women in Science and Engineering** program 2024, 2025

Mentoring Experience

- Served as a **teaching assistant** for **2 semesters**, for Linear Algebra and Classical & Quantum Physics 2024, 2025
- Selected in a team of **35+ mentors** from **175+ applicants** to provide academic guidance to 12 sophomores 2024
- One of **19** pre-final year mentors, selected from **1400+ peers** to provide mentorship to 12 freshmen 2024

Extracurricular Activities

- Appointed **Junior Consultant** at IITB's **Core Consulting Group** in recognition of strong communication skills 2023
- Secured **2nd place** in the Annual Freshers **Debate Open**, competing against **1400+** participants 2022
- Participated in the **State level Basketball tournament** of Tamil Nadu and represented the Chennai District 2018
- Awarded with distinction grade in **Level 5 Keyboard Certification**, by the **Trinity College London** 2018